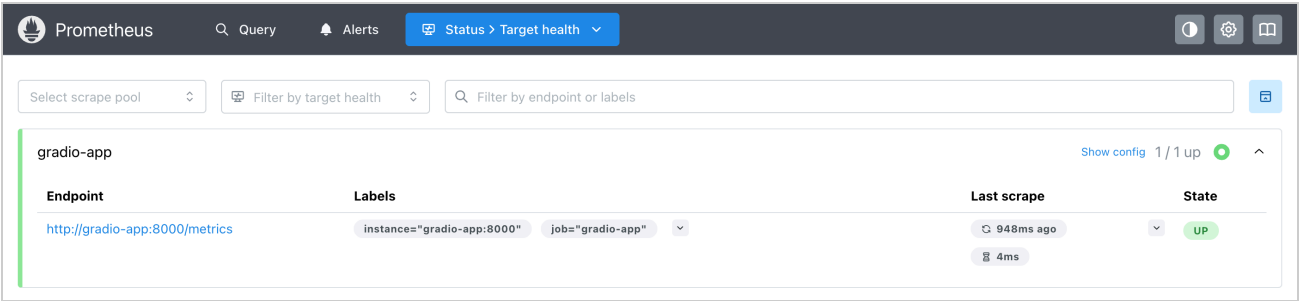
Module 8: Deployment and Monitoring

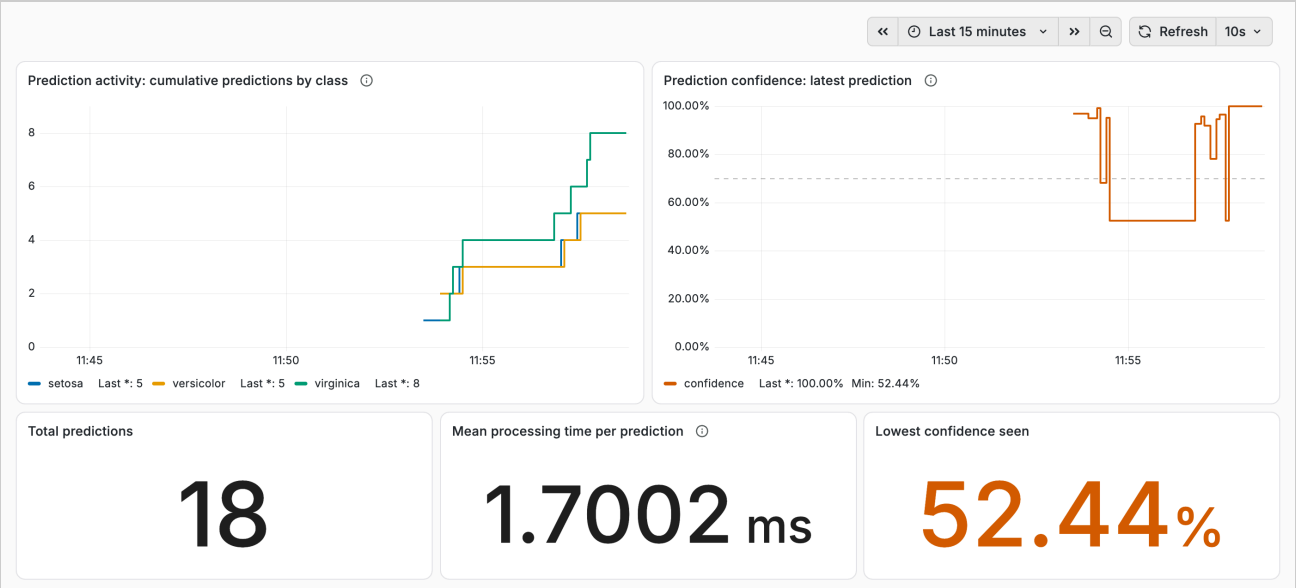
Mustafa Sameen · EGN 6216 · November 9, 2026

1. Prometheus target



Prometheus Target health page. The gradio-app target at http://gradio-app:8000/metrics is UP (1/1 up) and is scraped every 5 seconds.

2. Grafana dashboard



Dashboard after 18 test predictions made through the Gradio app. Both main panels show data.

3. What each panel measures

- **Prediction activity.** The running total of predictions for each species (setosa, versicolor, virginica). It plots the counter gradio_predictions_total summed by class. Each step up is one prediction of that class.
- **Prediction confidence.** The model's probability for the species it chose on the most recent prediction. It plots the gauge gradio_prediction_confidence from 0 to 100%. The dashed line at 70% marks a low-confidence level.
- **Total predictions.** The sum of all prediction counts (18 at the time of the screenshot).
- **Mean processing time per prediction.** The total time spent in the prediction function divided by the number of calls (request_processing_seconds_sum / request_processing_seconds_count).
- **Lowest confidence seen.** The minimum confidence recorded in the last 30 minutes.

4. Observation from the test predictions

Most flowers with typical measurements for their species were classified with 92 to 100% confidence. Confidence fell as the inputs moved toward the boundary between versicolor and virginica. (5.9, 3.0, 5.1, 1.8) gave virginica at 78% and (6.0, 2.7, 5.0, 1.6) gave virginica at 68%. The two most borderline inputs, (6.3, 2.5, 4.9, 1.5) and (6.2, 2.8, 4.8, 1.8), were labeled virginica with only about 52% confidence. These are the two deep drops below the dashed line in the confidence panel, and they set the lowest confidence seen (52.44%). The model is least sure exactly where the two species overlap in petal size.

Processing was fast throughout. The mean time per prediction was 2.4 ms after the first 10 predictions and 1.7 ms after 18.